

# Abdullah Khaled

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🔗 [akhaled247](https://github.com/akhaled247)

## Education

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**Wakeland High School**, High School Diploma – Frisco, TX Aug 2023 – May 2027

- Unweighted GPA: 4.0
- 18 AP courses by graduation (completed and planned)

**Frisco ISD Career and Technical Education Center**, Project Lead The Way – Frisco, TX Aug 2023 – May 2027

- Frisco ISD Career and Technical Education Center Project Lead the Way (PLTW) Nationally Distinguished Engineering Program
- Courses: PLTW Introduction to Engineering Design (9th grade), PLTW Principles of Engineering (10th grade), PLTW Digital Electronics (11th grade), PLTW Engineering Design and Development (12th grade)
- Engineering Portfolio: [akhaled247.github.io/pltw-engineering](https://akhaled247.github.io/pltw-engineering)

## Technical Experience

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**Programming Lead, Chief Technical Advisor**, ITKAN Robotics (FRC 9128) – Frisco, TX June 2024 – present

- Developed components of competition robot software stack integrating localization, path planning, telemetry, and mechanism control in Java
- Built publish-subscribe telemetry and motor logging framework for real-time robot diagnostics and performance analysis
- Led development of robot state estimation and localization system using sensor fusion of AprilTag vision and swerve-drive odometry, achieving <2 in positional error
- Implemented finite-state machine control architecture with Dijkstra-based path planning using time-weighted graph
- Used WPILib Java mechanism simulation to optimize telescoping arm geometry and motion constraints
- Developed self-paced robotics programming curriculum and training website used by 20+ FRC and FTC students and reaching 1000+ impressions worldwide ([mantik.netlify.app](https://mantik.netlify.app))
- Advanced team to Texas State Championship playoffs (89 teams)
- Robot codebase and documentation: [akhaled247.github.io/first/reefscape](https://akhaled247.github.io/first/reefscape)

**Programming Mentor**, ITKAN Robotics of ICF (FTC 26357, 28931) – Frisco, TX Sept 2024 – Feb 2026

- Implemented Kalman-filtered state estimation combining AprilTag vision and wheel odometry for real-time robot localization and automated turret targeting
- Mentored all-rookie FTC team through full engineering design cycle from prototyping to competition
- Mentor refugee students in FLL workshops and lead weekly summer training for FTC and FRC students using self-developed OARobotics curriculum ([akhaled247.github.io/personal/oarobotics](https://akhaled247.github.io/personal/oarobotics))

**Co-Founder, President**, Wakeland High School Wolverine Robotics (FTC 33791) – Frisco, TX Aug 2024 – present

- Co-founded Wakeland robotics club, growing membership from 0 to 50+ students in the first year
- Secured funding and constructed competition-ready robot in under 2 weeks
- Teach CAD software (Onshape) and Java programming through monthly technical workshops
- Manage club operations including mission development, curriculum design, and budget management

**Math and Reading Instructor**, Gideon Math and Reading Center – Frisco, TX

June 2024 – present

- Instruct K-11th grade students in mathematics (Algebra, Geometry), reading, and grammar
- Analyze student performance data to identify learning gaps and develop personalized plans
- Evaluate reading material difficulty and adjust instruction to match target comprehension levels

## Projects

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**OralVision – The Conrad Challenge**

Oct 2025 – present

- Designed oral disease detection system using HIPAA-compliant federated learning with TensorFlow Federated, enabling distributed model training without sharing patient data
- Used Onshape 3D CAD software to design and 3D print the device, reducing manufacturing costs
- Conrad Challenge Innovator Award and Alternate Finalist; Diamond Challenge Semi-Finalist
- Patent application in progress with the United States Patent and Trademark Office (USPTO)

**WatchFall – Samsung Solve for Tomorrow**

Sept 2025 – present

- Designed fall-detection device using Raspberry Pi Pico with on-device TensorFlow Lite inference
- Developed lightweight ML model for real-time motion tracking and fall detection with buzzer alert system
- Engineered hybrid dataset combining public and custom annotated data, reducing false positive rate

## Skills

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**Software & Data:** Python, Java, JavaScript, R, Google Sheets

**Robotics, ML & Vision:** TensorFlow, TensorFlow Federated, OpenCV, SolvePnP, Kalman Filtering, Finite State Machines, Mechanism Simulation, Control Theory, Federated Learning

**CAD, Electronics & Prototyping:** Fusion 360, Onshape, NI Multisim, KiCAD, EasyEDA, Soldering

**Web & Productivity:** HTML, CSS, Jekyll, Google Workspace, Microsoft Office

## Leadership & Activities

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**Vice President**, Math National Honor Society – Wakeland High School, Frisco, TX

Sept 2023 – present

- Led organization as Vice President (2024-25) and Treasurer (2023-24), managing budget for 100+ members
- Designed Math Lab Program website using Google Sites and coordinated peer tutoring, attaining over 20% member participation

**Secretary, Co-Founder**, Muslim Student Association – Wakeland High School, Frisco, TX

Sept 2024 – present

- Co-founded MSA as Secretary, establishing organizational structure and creating digital promotional media

**Lead Math and English Instructor**, Ma'ruf Project Taleem – Dallas, TX

Sept 2023 – present

- Teach Math and English to refugee elementary and middle school students, developing differentiated curriculum

## Honors & Awards

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- (2026) OralVision: Conrad Challenge Innovator Award & Alternate Finalist, Diamond Challenge Semifinalist
- (2026) FTC 33791: Event Winner, Control Award Winner (FiT-North Dallas Semi-Regional Championship, 25 teams), Control Award Winner (FiT-North U-League Tournament)
- (2025) FRC 9128: FIRST® Leadership Award Semifinalist, Event Winner (FIRST® in Texas Victoria, 25 teams); Autonomous Award (FiT Plano, 31 teams); Creativity Award (FiT Amarillo, 36 teams)
- (2025) FTC 26357: Finalist Alliance Captain (FiT-North McKinney Qualifier, 19 teams)
- (2025) Collegeboard: AP Scholar with Distinction, National Recognition Program: School Recognition Award